INCH-POUND
MIL-DTL-24643/65B
27 November 2023
SUPERSEDING
MIL-DTL-24643/65A
28 January 2010

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 150 VOLTS, TYPE LS3C179DT (INCLUDING VARIATION LS6C179DT)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable can be manufactured in two variations: LS3C179DT (three coaxial cables) and LS6C179DT (six coaxial cables).

Construction (Non-Watertight)

First	-	Solid copper conductor, coated or uncoated, in accordance with ASTM B3 or ASTM B33, 0.0122 inch diameter.
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation, 0.056 inch nominal diameter. Wall thickness may vary from nominal as necessary in order for completed cable to meet the specified electrical requirements.
Fourth	-	Light bonded aluminum foil/polyester tape shield applied with 25 percent minimum overlap. Foil side in contact with braid shield. 0.001-inch-thick aluminum foil.
Fifth	-	Braided shield of 38 AWG tin-coated copper, 95 percent minimum coverage.
Sixth	-	An optional binder.
Seventh	-	Cross-linked polyolefin jacket. Component diameter 0.100±0.005 inch. Standard identification code shall be applied on component jacket by method 1 or 3. If identification method 1 is used, the jacket color shall be black.
Eighth	-	The required number of coaxes (see table I) shall be cabled together with a lay in accordance with 3.4.4 of MIL-DTL-24643. Fillers may be used as necessary to make a firm, well-rounded assembly. A single coax shall not be used as a central core for the cable. Cabling sequence shall be consecutive.
Ninth	-	An optional binder.
Tenth	-	Braided shield of 36 AWG tin-coated copper, 90 percent minimum coverage.
Eleventh	-	An optional binder.

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Twelfth - Cross-linked polyolefin jacket. See table I for thickness.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

TABLE I. Details.

Military part no. M24643/65	Type and size	Number of coaxes	Cable jacket thickness min. avg. (inch)	Overall diameter (inch)		Cold bend mandrel diameter max. (inches)
				min.	max.	
-01UO	LS3C179DT	3	0.047	0.335	0.365	4.0
-02UO	LS6C179DT	6	0.050	0.425	0.455	5.0

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	80.0
Voltage withstand (volts, root mean square, min.)	
Conductor to shield	1,500
Component shield to all other shields	1,000
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to shield	500
Component shield to all other shields	100
Jacket flaws	No failure
Conductor and shield continuity	No failure

Group A:

Visual and dimensional	No failure
Characteristic impedance, at 1 megahertz (ohms) (Time Domain Reflectometer method in accordance with characteristic impedance of MIL-DTL-17)	75±5

Group B:

Thermoset proof test (percent, max.)	
Insulation	50
Jacket (component and overall) (when tested at 392 °F [200 °C])	50
Cold bending (cable) (see table I for mandrel diameter)	No failure

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Bending endurance at 77±3.6 °F (25±2 °C]) (cycles, min.)	25
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (component)	
Tensile strength (lb/in^2, min.)	1,000
Elongation (percent, min.)	100
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Group C:	
Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	80
Elongation (percent of unaged, min.)	80
Jacket (components)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Shrinkage	No failure
Heat distortion (percent of unaged, max.)	30
Permanence of printing (component jacket, method 1 only) (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Shield (conformance to material, construction, and coverage)	No failure
Surface transfer impedance	
Milliohms per meter (max.)	Not applicable
Electromagnetic pulse (EMP) response (dB, min.)	Not applicable

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Group D:

Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Dimensional stability (cable) (194±3.6 °F [90±2 °C]) (test in accordance with MIL-DTL-17)	
Inner conductor from coax core (inch, max.)	0.062
Inner conductor from coax jacket (inch, max.)	0.125
Abrasion resistance (overall jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket (component and overall)	2
Fillers	2
Insulation	18
Shield insulation (component jacket)	2
Halogen content (percent, max.)	
Jacket (component and overall)	0.2
Fillers	0.2
Insulation	0.2
Immersion (overall jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket (component and overall)	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket (component and overall)	5
Fillers	5
Insulation	1.5
Durometer (overall jacket) – type A (hardness, min.)	80
Weathering (overall jacket)	No failure
Electrical moisture absorption	No failure

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UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LS3C179DT and LS6C179DT	500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-081)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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